b) To draw power angle characteristics

Assume an induced emf of 120% of terminal voltage

$E_f = \frac{231 \times 120}{100} = 277.2V$

P_1= power due to field excitation (excitation power)

P_2=Reluctance power (power due to saliency)

$P_1 = \frac{3E_f V \sin\delta}{X_d} =$ __________W

$P_2 = \frac{3V^2(X_d - X_q)\sin 2\delta}{2X_d X_q} =$ __________W

Resultant power P = $P_1 + P_2$= __________W

TABULATION – Power Angle Characteristics

Load Angle δ	P_1 watts	P_2 watts	P watts
-180			
-150			
-135			
-120			
-90			
-60			
-45			
-30			
0			
30			
45			
60			
90			
120			
135			
150			
180			